

Tyzor® Catalysts for Polyols and Polymers

Application Guide



POLYMER	PRODUCT	APPLICATION NOTES
Polyester Polyol	Tyzor® TE	– Produces clear, colorless polyols. – Provides moderate reaction rates with higher rates at lower temperatures. – Provides good water stability and minimal hydrolysis reactions. – High yield; easy work-up; low toxicity; and low catalyst concentration.
	Tyzor® 9000	– The reaction components to prevent undesired pre-reactions. transesterifications run at low temperatures < 100 °C. Esterifications (e.g. polyester preparation) need temperatures of > 180 °C. – Produces clear, colorless polyols. – Provides fast reaction rates with higher rates at lower temperatures.
	Tyzor® LA	– Is a water based titanium chelate that is stable in water. – High yield, easy work up – For carboxylic acid direct esterification particularly when water concentrations in the raw materials are too high. – Fatty acid ester exchange or transesterification is another reaction that is catalyzed by TYZOR® LA
	Tyzor® TnBT	– Good reaction rates in polyester polyols. – Provides no water tolerance and is susceptible to hydrolysis that may increase maintenance downtime, slow reaction rates and produce haze.
	Tyzor® TPT	– Fast-acting titanate catalyst. – Good overall reaction-rate performance and excellent clarity and color. – High yield, easy work up – Transesterifications can be run at low temperatures < 100 °C. Direct esterifications of carboxylic acids need temperatures of > 180 °C.
	Tyzor® TOT	– High yield, easy work up, low catalyst concentrations – Transesterifications run at low temperatures < 100 °C. Esterifications (e.g. plasticizer preparation) need temperatures of > 180 °C. – Acts as Lewis acid giving fast catalysis under mild conditions in many applications such as esterification or transesterification, polycondensation etc
plasticizers	Tyzor® TPT	– High product quality, high yield – Easy processibility – Catalysis of the esterification reaction step at > 180 °C. – Addition of low amounts
	Tyzor® TnBT	– High product quality, high yield – Easy processibility – Catalysis of the esterification reaction step at > 180 °C. – Addition of low amount
Acrylate esters	Tyzor® TPT	– (poly)addition reactions to produce (meth-)acrylic esters. – Fast-acting titanate catalyst. – Produces high reaction rates, excellent clarity and low color.
	Tyzor® TNBT	– (poly)addition reactions to produce (meth-)acrylic esters. – Fast-acting titanate catalyst. – Produces high reaction rates, excellent clarity and low color.
	Tyzor® AA 75	– (poly)addition reactions to produce (meth-)acrylic esters. – Hydrolytically stable titanate catalyst. – Produces high reaction rates, excellent clarity and low color.

POLYMER	PRODUCT	APPLICATION NOTES
PBT	Tyzor® 422	– Excellent thermal properties for resin – High reactivity for polycondensation with SSP rates similar to antimony – Low-residue technology produces brighter, lower-haze PET than conventional catalysis – Stable and easy to handle
	Tyzor® 436	– Hydrolytically stable titanate catalyst. and easy to handle – Ultra-low-residue technology produces brighter, haze-free PBT than conventional catalysis – Minimal TiO₂ cluster and extending filter lifetimes – Greatly improved polymer thermal/thermo-oxidative stability for fiber.
	Tyzor® PC 64	– Hydrolytically stable titanate catalyst. – Reduced Catalyst Loading – Process Optimization with higher rates at lower temperatures. – Minimal TiO₂ cluster and extending filter lifetimes
	Tyzor® TnBT	– Good reaction rates in PBT. – High Ti-content and reactivity – Higher efficiency – Produces high reaction rates, excellent clarity
PET	Tyzor® TnBT	– High yield, easy work up, High Ti-content and reactivity – Fast-acting titanate catalyst. – Produces high reaction rates, excellent clarity and low color.
	Tyzor® PC 64	– Process Optimization with higher rates at lower temperatures. – Hydrolytically stable titanate catalyst. – Reduced Catalyst Loading – Improved Quality of Bottle Resin
	Tyzor® ACtivate™ 422	– Excellent for bottle and film resins. – Hydrolytically stable titanate catalyst. – Produces high reaction rates, excellent clarity and low color. – Tyzor® ACtivate™ 422 can increase productivity and significantly reduce manufacturing costs
	Tyzor® ACtivate™ 436	– Stable and easy to handle – Hydrolytically stable titanate catalyst. – Excellent thermal properties for fiber – Ultra-low-residue technology produces brighter, haze-free PET than conventional catalysis
Polycarbonate	Tyzor® TPT	– Good overall reaction-rate performance and excellent clarity and color. – High yield, easy work up, High Ti-content and reactivity.
	Tyzor® TnBT	– Good overall reaction-rate performance and excellent clarity and color. – High yield, easy work up, High Ti-content and reactivity.
	Tyzor® BTP	– High yield, easy work up, High Ti-content and reactivity. – Provides good water stability and minimal hydrolysis reactions.